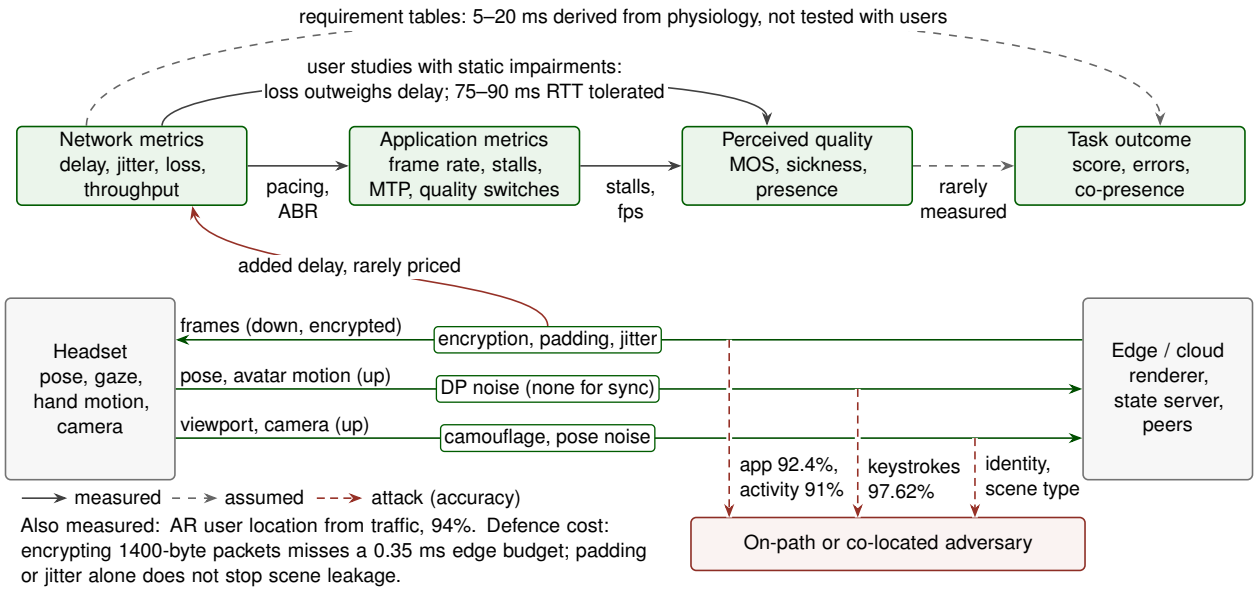


Figure 1: Measurement chain and attack surfaces of a networked XR session. Solid arrows are links that user or traffic studies have measured, dashed arrows links that are assumed; red taps show measured attack accuracies, and defences sit on the flows where they add delay that has rarely been priced against the frame deadline.

1 Assuring Experience and Trust: QoE, Security and Privacy

The previous sections optimise parts of the loop against targets. This section asks whether the result is good enough for its users, and safe for them. Experience and trust share three properties. First, they belong to the whole loop, not to any one component. A frame arrives late because of the encoder, the scheduler, the pacing of the transport and the reprojection on the headset together, and a pose leaks whether it travels to a renderer, a cache or a peer. Second, both are judged at the human end: by what a user perceives, and by what an adversary learns about that user. Third, both are usually inferred from proxies. Network metrics stand in for experience, and attack accuracy on a captured dataset stands in for risk. The section therefore has two halves. On experience, it asks what users actually weigh and how experience and traffic are modelled; on trust, it asks what XR traffic reveals and what defences cost. Figure 1 draws the measurement chain, the attack surfaces and the points where defences add delay.

1.1 What users weigh

Users of remote-rendered XR weigh loss and stalls above added delay of tens of milliseconds. Four platforms agree. On ALVR, round-trip times (RTT) up to 90 ms were hardly noticed, while 8% loss and a 35 Mbps cap degraded experience [1]. With CloudXR (N=30), QoE fell at RTT of 75 ms or more, at loss above 6%, or at 25 ms combined with 4% loss [2]. Loss also outweighed delay for presence and emotion on Quest 3 (N=16) [3], and in social XR an RTT up to 77 ms had no significant effect while jitter did [4].

The headset separates two latencies. Rotation latency, the motion-to-photon (MTP) delay of turning the head, is served locally by reprojection; network RTT reaches only interaction latency, the delay of aiming, grabbing or seeing a peer react (**??**). Isolated stalls of up to four frames went unnoticed, and reprojection helped from eight frames (N=29) [5]. This explains a contested budget. Requirement papers derive 5–10 ms RTT and 20 ms interaction latency from physiology [6], a factor of four to ten below what users tolerated. Reprojection plausibly reconciles the two numbers, but no study has varied RTT and latency compensation together, so the budget that edge placement should meet remains open.

The evidence also separates the outcomes that the network moves from those it does not. Viewport-adaptive delay lowered perceived quality but barely changed presence and left sickness almost untouched (20 observers) [7]. Fewer than 22% of viewers reported sickness for AR point clouds [8], and it was “not an issue” for 360-degree video [9]; content motion and display, not the network, seem to drive it. For volumetric content, frame rate dominated in a 102-participant study [10], and two quality levels were indistinguishable at 5 m on a HoloLens 2 (N=32) [8]. Task performance, the outcome that

Table 1: Families of QoE models for networked XR. Each family is limited by what it observes and by the ground truth it was validated against.

Family	Observes	Validated against	Limit
Parametric standard (P.1203, P.1204.3 tiled)	Bitrate, resolution, stalls, quality switches	Ratings of short streaming stimuli	No interaction latency; 10 s stimuli
Objective media metrics (PSNR, VMAF, colour PSNR)	Decoded pixels or points against a reference	Ratings under coding impairments	Blind to timing; content dependent
Learned estimators	Network and application metrics, game state	Ratings on a few contents in one lab	Fitted and tested on the same small set
Network-side estimators	Encrypted traffic, radio and in-network counters	Parametric indices or activity labels	Never checked against users for interactive XR
Requirement tables	Human perception limits	Physiology, not networked user studies	4–10$\times$ stricter than tolerated RTT

matters for training or teleoperation, is rarely measured beside ratings. Group effects are tested almost only in pairs with the impairment on one link [11]. Players blamed their opponent for their own delay [12], and loading avatar and world models congested shared networks as groups grew [13]; who is delayed in a larger group, and how the others judge it, is unknown (**??**).

These results share a study design: *static impairments*. Every user study applies constant RTT, independent loss or throttling with tools such as NetEm. Real 4G and 5G measurements enter only after the fact, as inputs to models fitted under constant impairments [14]. No study has *replayed measured network dynamics* to users. Wireless loss is bursty and correlated, through aggregation, scheduling and handover, so thresholds learned under independent loss may not transfer.

1.2 How experience and traffic are modelled

A QoE model maps what can be observed to what a user would report. Four families differ in what they observe and what they were validated against (Table 1). *Parametric standard models* extend ITU-T P.1203 and P.1204.3 to tiled streaming [15, 16], with P.919 as the test method for short stimuli. *Objective media metrics* are established for coding impairments: PSNR tracks MOS for 360-degree video [17], VMAF correlated best on a public dataset [18], and colour PSNR reached $r = 0.84$ for point clouds, depending on content [8]. Neither family sees interaction latency. *Learned estimators* add it but fit locally. A cloud VR gaming model reached a Pearson correlation of 0.93 and, when used to adapt encoding, raised interaction quality by 1.20 MOS [19]; regressions on the CloudXR data reached a mean absolute error of 0.14 [14]. *Network-side estimators* infer quality from what an operator sees: radio traces, encrypted 360-degree traffic from drive tests [20], inter-packet gaps in a programmable data plane [21], or an LSTM on an O-RAN testbed [22]. They are validated against formulas, not people. PERCEIVE predicted a parametric QoE index within 4%, and its authors state that the index was never checked against ratings [23].

All four families lack *temporal pooling*. P.919 uses 10 s stimuli, while presence needs multi-minute tasks [25], and no model pools sub-second network events into the QoE of a session.

Traffic is characterised in two ways. *Measurements of commercial platforms* show that the application stack, not the link, often sets what users see. WebRTC pacing in 5.56 ms batches set frame delay at 6–21 ms over Wi-Fi 6 with an RTT of 1.5 ms [26]. Under 27 Mbps throttling, adaptive bitrate kept cloud VR frame loss at or below 5.12% where a fixed 40 Mbps stream lost 32.51% [27]: adaptation converts a throughput shortfall into image quality rather than lost frames. Horizon Workrooms used 2–3 Mbps downlink and 0.6 Mbps uplink [28], two orders of magnitude below the per-user rates in metaverse visions (**??**). *Statistical traffic models* capture constant-bitrate video well: frame size follows bitrate over frame rate, and frame intervals follow the frame period with light-tailed or log-logistic marginals [27, 29]. They are open loop, however. The 3GPP baseline underestimated effective VR bitrate by 25% [30], and the open RiftCat model matched mean but not 95th-percentile delay [29]. Recent models add head motion [31] or session states [32], but none couples user motion with the stack's rate adaptation and congestion control. Schedulers evaluated on such models misjudge capacity

in the tail.

1.3 What XR traffic reveals

XR transmits the body. Pose, gaze, viewport requests and camera frames are control signals for the loop and, at the same time, behavioural and biometric data. Encryption hides their content but not their size, timing or recipients, and four attack surfaces follow.

Traffic metadata reveals place and activity, and the result is established across platforms and groups. Traffic patterns of a mobile AR application located users with 94% accuracy [33]. On Meta Quest Pro, off-the-shelf classifiers identified applications with 92.4% accuracy and activities with 91%, from less than 10 minutes of traffic per application [34]. *Motion as a biometric* is the sharpest channel. Avatar motion received by an adversary's own client in multi-user VR yielded 97.62% of typed keystrokes across four applications [36], and Wi-Fi channel state alone yielded 69.75% [37]. State synchronisation, which broadcasts head and hand poses to every peer, thus becomes a leakage channel by design (**??**). *Viewport requests* identify users because prediction-driven tile requests follow individual head motion (**??**); viewpoint leakage persists even under federated prediction once prediction errors or QoE are uploaded [38]. *The camera uplink* of offloading AR glasses reveals the type of physical scene through packet sizes and timing [39], so the split chosen in **??** is also a privacy decision. Finally, the provider is an observer too: about 70% of Oculus data flows were not properly disclosed [40].

1.4 Defences and their price

Every defence spends a resource the loop is short of: delay, bandwidth or prediction accuracy, and the price is rarely measured.

Encryption is cheap for small packets and expensive at the tightest budgets. Encrypting 1400-byte packets did not fit a 0.35 ms edge budget, while 256-byte packets did [41]. Selective attribute-based encryption cut encryption and decryption time by up to 50% and 80% and kept volumetric content cacheable [42]. Secured VR streaming under a flood held about 30–32 ms in simulation [43]. *Obfuscation* hides metadata at a cost in bandwidth or quality. Padding or jitter alone did not stop scene leakage, and combined defences conflicted with QoS [39]. Camouflaged tile requests need more resources and degrade prediction [45]. Noise remapping reduced the QoE loss of privacy noise by 55–100%, to 0–3% at maximum privacy [46], and client-side pose noise lowered re-identification on real 6DoF traces [47]; in this line the QoE loss is computed from prediction accuracy, and no user has been asked.

Authentication schemes for metaverse avatars and tele-surgery are proven in BAN logic, AVISPA or Real-or-Random models and costed in operation counts [48, 49]. None is measured during an edge or platform hand-off, where re-authentication falls on the interaction path (**??**). *Federated and edge processing* keeps raw data local but does not by itself protect it: federated learning for VR biometric authentication reached below 10% accuracy [51]. *Detection* rarely sees XR traffic. "Metaverse" intrusion detectors report 96–99% accuracy on IoT and enterprise datasets that contain no XR flows [52]; the exceptions are line-rate detection in Tofino switches at up to 99% accuracy with under 5% of switch resources [53] and the first labelled metaverse attack dataset [54]. *Blockchain* is contested: a 10-user pilot kept 88 FPS at 142 transactions per second [55], while most ledger frameworks report no latency at all. What decides the question is whether a ledger write sits on the interaction path.

The privacy–QoE trade-off is therefore modelled, not perceived. No testbed study reports a defence's latency together with QoE or frame rate. A defence should be reported as a point on a privacy–QoE plane: attacker accuracy against added 99th-percentile frame latency and bandwidth, on a real split-rendering pipeline.

Open problems

P1. Do network-side estimators predict what XR users report? Estimators target parametric indices, and none addresses interactive remote rendering. A public benchmark pairing encrypted packet traces, client frame logs and per-minute ratings for cloud VR would let estimators be trained and compared against people.

P2. Do thresholds from static impairments hold under real wireless dynamics? Users have only seen constant RTT and independent loss. A study replaying measured Wi-Fi and 5G traces, with reprojection

on and off, would fix the network latency budget for remote-rendered XR.

P3. How do sub-second events pool into session QoE, and into group QoE? No model links bursts of stalls or delay to a multi-minute session, and groups beyond pairs are untested. Scripted burst schedules and asymmetric impairments in groups of four to six would answer both.

P4. Can a closed-loop traffic model reproduce the tail? Models fitted to one stack ignore head motion and rate adaptation, and err at the 95th percentile. A generator coupling pose, adaptation and pacing, validated against several real headsets, would make scheduler studies credible.

P5. What does a defence cost against the frame deadline? Measuring TLS, DTLS and QUIC with padding, bucketing and jitter on split rendering, as attacker accuracy against 99th-percentile frame latency, would give providers a price; the same holds for re-authentication during hand-off.

P6. Can state synchronisation share avatars without sharing biometrics? Poses broadcast to every peer reveal keystrokes, and no synchronisation protocol limits this. A protocol that perturbs or coarsens remote poses, tested for keystroke inference, co-presence and sickness, would close the channel.

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